

Number System

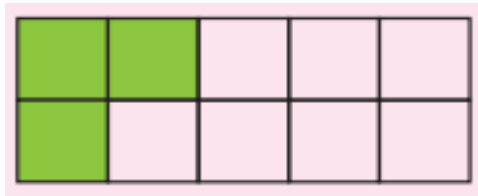
Choose the Appropriate Answer

1. Find the Fractional representation of the following picture



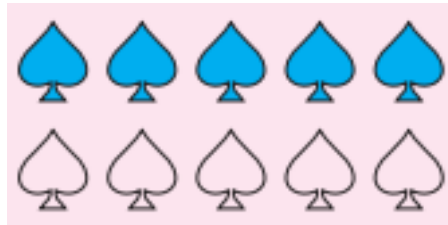
- (a) $\frac{4}{9}$ (b) $\frac{1}{2}$ (c) $\frac{3}{8}$ (d) $\frac{5}{8}$
Ans: (b) $\frac{1}{2}$

2. Find the decimal form of the given picture



- (a) 0.03 (b) 0.3 (c) 0.003 (d) None
Ans: (b) 0.3

3. Find the decimal form of the given picture



- (a) 0.2 (b) 0.4 (c) 0.5 (d) 0.1
Ans: (c) 0.5

4. The place value of 3 in 85.073 is _____
(a) Tenths (b) Hundredths (c) Thousands (d) Thousandths
Ans: (d) Thousandths

5. To convert grams into kilograms, we have to divide it by
(a) 10000 (b) 1000 (c) 100 (d) 10
Ans: (b) 1000

6. The decimal representation of 30 kg and 43 g is _____ kg.
(a) 30.43 (b) 30.430 (c) 30.043 (d) 30.0043
Ans: (c) 30.043

7. A cricket pitch is about 264 cm wide. It is equal to _____ m.
(a) 26.4 (b) 2.64 (c) 0.264 (d) 0.0264

Ans: (b) 2.64

8. $3 + \frac{4}{100} + \frac{9}{1000} = ?$

- (a) 30.49 (b) 3049 (c) 3.0049 (d) 3.049

Ans: (d) 3.049

9. $3 / 5 =$
(a) 0.06 (b) 0.006 (c) 6 (d) 0.6

Ans: (d) 0.6

10. The simplest form of 0.35 is
(a) $35 / 1000$ (b) $35 / 10$ (c) $7 / 20$ (d) $7 / 100$

Ans: (c) $7 / 20$

11. 0.009 is equal to
(a) 0.90 (b) 0.090 (c) 0.00900 (d) 0.900

Ans: (c) 0.00900

12. $37.7 \text{ ----- } 37.7$
(a) = (b) < (c) > (d) $\neq$

Ans: (a) =

13. $78.56 \text{ ----- } 78.57$
(a) < (b) > (c) = (d) $\neq$

Ans: (a) <

14. Between which two whole numbers 1.7 lie?
(a) 2 and 3 (b) 3 and 4 (c) 1 and 2 (d) 1 and 7

Ans: (c) 1 and 2

15. The decimal number which lies between 4 and 5 is _____
(a) 4.5 (b) 2.9 (c) 1.9 (d) 3.5

Ans: (a) 4.5

16. The place value of 3 in 85.073 is _____.
(a) Thousandths (b) Hundredths (c) Tenths (d) None

Ans: (a) Thousandths

17. To convert grams into kilograms, we have to divide it by
(a) 10 (b) 100 (c) 1000 (d) 10000

Ans: (c) 1000

18. The decimal representation of 30 kg and 43 g is _____ kg.
(a) 30.43 (b) 30.043 (c) 30.0043 (d) 3.043

Ans: (b) 30.043

19. A cricket pitch is about 264 cm wide. It is equal to _____ m.
(a) 0.264 (b) 0.0264 (c) 2.64 (d) 26.4

Ans: (c) 2.64

20. The height of a person is 165 cm. Express this height in metre.
(a) 0.165 (b) 1.65 (c) 16.5 (d) 0.0165

Ans: (b) 1.65

21. $3 + (5 / 10) + (3 / 100) + (4 / 1000) =$
(a) 3.534 (b) 0.3534 (c) 35.34 (d) 0.03534

Ans: (a) 3.534

22. $40 + 6 + (7 / 10) + (2 / 100) + (6 / 1000) =$
(a) 467.26 (b) 46.726 (c) 406.726 (d) 4.6726

Ans: (b) 46.726

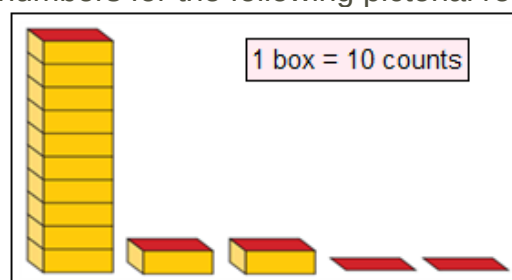
23. Find the place value: 54.58
(a) Thousandths (b) Hundredths (c) Tenths (d) None

Ans: (c) Tenths

24. Find the place value: 0.855
(a) Thousandths (b) Hundredths (c) Tenths (d) None

Ans: (b) Hundredths

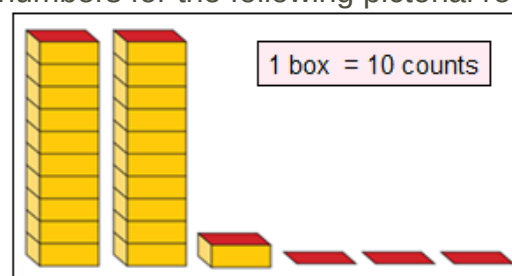
25. Write the decimal numbers for the following pictorial representation of numbers.



(a) 1022 (b) 122 (c) 1202 (d) None

Ans: (b) 122

26. Write the decimal numbers for the following pictorial representation of numbers.



(a) 2023 (b) 2103 (c) 213 (d) 2133

Ans: (c) 213

27. Find the equivalent decimal number

Tens (10)	Ones (1)	Tenths (1 / 10)	Hundredths (1 / 100)	Thousandths (1 / 1000)
4	9	6	5	5

- (a) 49.655 (b) 496.55 (c) 4. 9655 (d) None

Ans: (a) 49.655

28. Find the equivalent decimal number

Thousands (1000)	Hundreds (100)	Tens (10)	Ones (1)	Tenths (1 / 10)	Hundredths (1 / 100)
4	9	5	4	9	5

- (a) 495.495 (b) 49.5495 (c) 4954.95 (d) None

Ans: (c) 4954.95

29. Face value of the given number 53.61

- (a) 60 (b) 6 (c) 6 / 10 (d) None

Ans: (b) 6

30. Place value of the given number 263.271

- (a) Thousandths (b) Hundredths (c) Tenths (d) None

Ans: (c) Tenths

31. Face value of the given number 17.39

- (a) 900 (b) 90 (c) 9 (d) None

Ans: (c) 9

32. Place value of the given number 9.657

- (a) Thousandths (b) Hundredths (c) Tenths (d) None

Ans: (b) Hundredths

33. Place value of the given number 4972.068

- (a) 8 / 10 (b) 8 (c) 8 / 1000 (d) None

Ans: (b) 8

34. Which of the following is a proper fraction?

- (a) 7 / 4 (b) 19 / 4 (c) 14 / 5 (d) 4 / 11

Ans: (d) 4 / 11

4 / 11 is a proper fraction because $4 < 11$
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35. Which of the following is an improper fraction?

- (a) 2 / 7 (b) 1 / 2 (c) 2 / 3 (d) 73 / 10

Ans: (d) 73 / 10

36. The improper fraction $33/4$ in the form of a mixed fraction is

- (a) $8\frac{1}{4}$ (b) $4\frac{1}{8}$ (c) $3\frac{8}{4}$ (d) $4\frac{7}{8}$

Ans: (a) $8\frac{1}{4}$

$$8\frac{1}{4} = \frac{32+1}{4} = \frac{33}{4}$$

37. Which of the following is an equivalent fraction of $2/3$?

- (a) $4/9$ (b) $6/13$ (c) $8/11$ (d) $10/15$

Ans: (d) $10/15$

$$\frac{2}{3} = \frac{2 \times 5}{3 \times 5} = \frac{10}{15}$$

38. $1 - 1/5$ is equal to

- (a) $2/5$ (b) $3/5$ (c) $4/5$ (d) $1/5$

Ans: (c) $4/5$

39. $2 + 1/4$ is equal to

- (a) $7/4$ (b) $9/4$ (c) $5/4$ (d) $11/4$

Ans: (b) $9/4$

40. $1/2 + 1/3$ is equal to

- (a) $1/6$ (b) $5/6$ (c) $7/6$ (d) $11/6$

Ans: (b) $5/6$

41. Manish worked for $1/2$ an hour. Rishi worked for $1/4$ of an hour. For how much time did both work together?

- (a) 1 hour (b) $3/4$ (c) $5/4$ (d) $3/2$

Ans: (b) $3/4$

42. The length of a rectangular sheet of paper is $3/5$ cm and the breadth is $2/5$ cm. What is the perimeter of the paper?

- (a) 1 cm (b) 2 cm (c) 3 cm (d) 4 cm

Ans: (b) 2 cm

43. $2 \times 5/3$ is equal to

- (a) $7/3$ (b) $11/3$ (c) $13/3$ (d) $10/3$

Ans: (d) $10/3$

44. $7/5 \times 4$ is equal to

- (a) $21/5$ (b) $22/5$ (c) $28/5$ (d) $23/5$

Ans: (c) $28/5$

45. There are 40 students in a class. $1/2$ of the total number of students are Hindus. The number of Hindus is

- (a) 10 (b) 20 (c) 30 (d) 40

Ans: (b) 20

46. A water bottle contains 2 litres of water. Melena drank $\frac{1}{8}$ of water. How much water did Melena drink?

- (a) $\frac{1}{2}$ litre (b) $\frac{1}{8}$ litre (c) $\frac{1}{4}$ litre (d) None of these

Ans: (c) $\frac{1}{4}$ litre

47. $5 \div \frac{5}{2}$ is equal to

- (a) 1 (b) 2 (c) 5 (d) 3

Ans: (b) 2

48. $\frac{1}{3} \div \frac{5}{3}$ is equal to

- (a) $\frac{1}{3}$ (b) $\frac{5}{3}$ (c) $\frac{1}{5}$ (d) None of these

Ans: (c) $\frac{1}{5}$

49. 5 paise is equal to

- (a) Rs.0.5 (b) Rs.0.05 (c) Rs.0.005 (d) Rs.0.0005

Ans: (b) Rs.0.05

50. 150 g is equal to

- (a) 0.15 kg (b) 0.015 kg (c) 0.0015 kg (d) 0.00015 kg

Ans: (a) 0.15 kg

51. The place value of 2 in 2.38 is

- (a) 1 (b) 2 (c) 20 (d) 0.2

Ans: (b) 2

$$\text{Place value} = 2 \times 1 = 2$$

52. How much less is 10 km than 30.6 km?

- (a) 10.6 km (b) 20.6 km (c) 30.6 km (d) 10 km

Ans: (b) 20.6 km

53. Find the area of a rectangle whose length is 6.7 cm and breadth is 2 cm.

- (a) 13 cm^2 (b) 13.4 cm^2 (c) 13.8 cm^2 (d) 14 cm^2

Ans: (b) 13.4 cm^2

54. If $43m = 0.086$, then the value of m is

- (a) 0.002 (b) 0.02 (c) 0.2 (d) 2

Ans: (a) 0.002

$$\text{Given that, } 43m = 0.086 \text{ m} = 0.086 \text{ } 43 \text{ m} = 0.086 \text{ } 43 \text{ m} = 0.002$$

55. Fractions with different denominators are called _____ fractions.

- (a) Like (b) Unlike (c) Both (a) & (b) (d) None

Ans: (b) Unlike

$$\text{Fractions with different denominators are called, unlike fractions}$$

56. Which one of the following is smaller?

- (a) 2.03 (b) 2.301 (c) 0.2301 (d) 23.01

Ans: (c) 0.2301

57. 7 Rupees 7 paise can be written in rupees as

- (a) Rs.7.07 (b) Rs.7.70 (c) Rs.0.707 (d) Rs.770

Ans: (a) Rs.7.07

58. 5 cm in Km can be written as

- (a) 0.0005 (b) 0.00005 (c) 0.0005 (d) 0.05

Ans: (b) 0.00005

$1 \text{ cm} = 100,000 \text{ km}$ $5 \div 100,000 = 0.00005$
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59. The place value of 2 in 21.38 is

- (a) One (b) Tens (c) Tenth (d) Hundredth

Ans: (b) Tens

60. The product of 153.7 and 10 is

- (a) 1.53 (b) 15.37 (c) 153.7 (d) 1537

Ans: (d) 1537

61. The value of 2.71×5 is

- (a) 135.5 (b) 1355 (c) 13.55 (d) 1.355

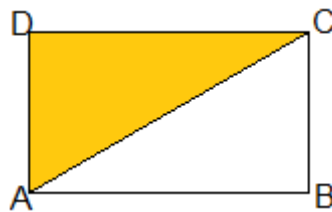
Ans: (a) 135.5

62. What is the sum of 5.300 and 3.250?

- (a) 8.550 (b) 85.50 (c) 5.6250 (d) 8550

Ans: (a) 8.550

63. What percentage of the adjoining figure is shaded?



- (a) 50% (b) 40% (c) 100% (d) None of these

Ans: (a) 50%

64. Which one of the following is greater?

- (a) 5.0 (b) 0.5 (c) 0.005 (d) 0.05

Ans: (a) 5.0

65. The value of $52.5 \div 10$

- (a) 5.25 (b) 0.525 (c) 525 (d) 5250

Ans: (a) 5.25

66. The value of $0.78 / 100$

- (a) 7800 (b) 0.0078 (c) 0.78 (d) 7.8

Ans: (b) 0.0078

67. The value of $26.3 / 1000$

- (a) 0.0263 (b) 0.263 (c) 2630 (d) 26.300

Ans: (a) 0.0263

68. The value of $7.75 / 0.25$

- (a) 31 (b) 0.0031 (c) 0.31 (d) 3.1

Ans: (a) 31

69. The equivalent rational number of $-6/5$ is

- (a) $-3/5$ (b) $12/10$ (c) $-12/5$ (d) None of these

Ans: (c) $-12/5$

70. Fill in the boxes with the correct symbol $\frac{5}{7} \square \frac{2}{3}$

- (a) $>$ (b) $<$ (c) $=$ (d) None of these

Ans: (b) $<$

71. Fill in the boxes with the correct symbol: $\frac{4}{5} \square \frac{-5}{7}$

- (a) $>$ (b) $<$ (c) $=$ (d) None of these

Ans: (a) $>$

72. Fill in the boxes with the correct symbol: $\frac{-7}{8} \square \frac{14}{16}$

- (a) $>$ (b) $<$ (c) $=$ (d) None of these

Ans: (b) $<$

73. Fill in the boxes with the correct symbol: $-\frac{5}{11} \square \frac{-5}{11}$

- (a) $>$ (b) $<$ (c) $=$ (d) None of these

Ans: (c) $=$

74. Fill in the boxes with the correct symbol: $\frac{1}{-3} \square \frac{-1}{4}$

- (a) $>$ (b) $<$ (c) $=$ (d) None of these

Ans: (b) $<$

75. Rewrite the rational number $-1\frac{8}{48}$ in the simplest form

- (a) $-9/24$ (b) $-3/8$ (c) $3/8$ (d) None of these

Ans: (b) $-3/8$

76. Rewrite the rational number $-24/72$ in the simplest form.

- (a) $12/-36$ (b) $6/-18$ (c) $1/-3$ (d) None of these

Ans: (c) $1/-3$

77. Rewrite the rational number $44 / 72$ in the simplest form.
 (a) $22 / - 36$ (b) $11 / - 18$ (c) $11 / 18$ (d) None of these
Ans: (c) $11 / 18$

78. Write the next rational number in the pattern: $\frac{-3}{5}, \frac{-6}{10}, \frac{-9}{15}, \frac{-12}{20}, \dots$
 (a) $12 / 25$ (b) $15 / 25$ (c) $- 15 / 25$ (d) None of these
Ans: (c) $- 15 / 25$

79. Find the value of $x : (8 / x) = (- 1 / 5)$
 (a) -5 (b) -40 (c) Any number (d) None of these
Ans: (b) -40

80. Find x such that $(7 / 3) = (- x / 6)$
 (a) -14 (b) -3 (c) $- 21$ (d) None of these
Ans: (a) -14

81. Find x such that $16 / x = - 4$
 (a) 4 (b) -4 (c) 2 (d) None of these
Ans: (b) -4

82. Find x such that $48 / x = 2$
 (a) 24 (b) -12 (c) -24 (d) None of these
Ans: (a) 24

83. Find the value of $(9 / 2) \times (- 4 / 3)$
 (a) 6 (b) -6 (c) 1 (d) None of these
Ans: (b) -6

84. Find the value of $(3 / - 5) \times (- 5 / - 3)$
 (a) 1 (b) 0 (c) $- 1$ (d) None of these
Ans: (c) -1

85. Find the value of $(3 / 10) \times (- 2 / 3)$
 (a) 5 (b) $1 / 5$ (c) $- 1 / 5$ (d) None of these
Ans: (c) $-1 / 5$

86. Find the value of $(- 6) / (2 / 3)$
 (a) -9 (b) 9 (c) $- 4$ (d) None of these
Ans: (a) $- 9$

87. Rational number $3 / 40$ is equal to:
 (a) 0.075 (b) 0.12 (c) 0.012 (d) None of these
Ans: (a) 0.075

88. A rational number between 3 and 4 is:
 (a) $3 / 2$ (b) $4 / 3$ (c) $7 / 2$ (d) None of these
Ans: (c) $7 / 2$

The sum of the numbers is $3 + 4 = 7$ $\Rightarrow$ The average is $\frac{7}{2} = 3.5$
 $\frac{7}{2}$ is greater than 3 and less than 4, it lies between the two numbers

89. A rational number between $\frac{1}{2}$ and $\frac{3}{4}$ is:

- (a) $\frac{2}{5}$ (b) $\frac{5}{8}$ (c) $\frac{4}{3}$ (d) $\frac{1}{4}$

Ans: (b) $\frac{5}{8}$

The first fraction becomes: $\frac{1}{2} = \frac{1 \times 4}{2 \times 4} = \frac{4}{8}$
 The second fraction becomes: $\frac{3}{4} = \frac{3 \times 2}{4 \times 2} = \frac{6}{8}$
 A rational number between $\frac{4}{8}$ and $\frac{6}{8}$ is $\frac{5}{8}$.

90. The multiplicative inverse of $\frac{3}{40}$ is:

- (a) 1 (b) 0 (c) Any number (d) None of these

Ans: (d) None of these

91. The additive inverse of $\frac{7}{5}$ is

- (a) 1 (b) 0 (c) $-\frac{7}{5}$ (d) $\frac{7}{5}$

Ans: (c) $-\frac{7}{5}$

92. Zero has _____ reciprocal.

- (a) 1 (b) 2 (c) 3 (d) no

Ans: (d) no

92. The numbers ----- and ----- are their own reciprocals

- (a) 1 and 0 (b) 1 and -1 (c) -1 and 0 (d) None of these.

Ans: (b) 1 and -1

93. The reciprocal of -5 is _____.

- (a) 5 (b) 1 (c) $-\frac{1}{5}$ (d) $\frac{1}{5}$

Ans: (c) $-\frac{1}{5}$

94. Reciprocal of $\frac{1}{x}$, where x is not equal to 0 is _____.

- (a) 1 (b) x (c) 0 (d) None of these

Ans: (b) x

95. The product of two rational numbers is always a _____.

- (a) Whole numbers (b) Integers (c) Natural numbers (d) Rational numbers

Ans: (d) Rational numbers

96. Which of the following fraction has numerator 5

- (a) $\frac{2}{5}$ (b) $\frac{5}{7}$ (c) $\frac{1}{5}$ (d) $\frac{1}{5}$

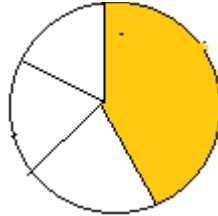
Ans: (b) $5/7$

97. Which of the following fraction has denominator 8.

- (a) $8/3$ (b) $1(3/8)$ (c) $1(8/3)$ (d) $3/8$

Ans: (d) $3/8$

98. What fraction does the shaded portion in the adjoining figure represents.



- (a) $5/2$ (b) $3/5$ (c) $2/5$ (d) $5/3$

Ans: (c) $2/5$

99. What type of the number is $-3/0$?

- (a) A positive rational number
(b) A negative rational number
(c) Either a positive or a negative rational number
(d) Neither a positive nor a negative rational number

Ans: (d) Neither a positive nor a negative rational number

The expression $-3/0$ is undefined because division by zero is not possible

100. The decimal number which lies between 4 and 5 is?

- (a) 4.5 (b) 2.9 (c) 1.9 (d) 3.5

Ans: (a) 4.5